2.3 Probability 2

1. Two fair dice are rolled. What is the conditional probability that at least one lands on 6 given that the dice land on different numbers?

2. If two fair dice are rolled, and the sum of the numbers is recorded.
   a) What is the probability that the first one lands on 6 given that the sum of the dice is 8?
   b) What is the probability that the first one lands on 6 given that the sum of the dice is 12?

3. Events A and B are independent, and $P(A) = \frac{1}{3}$ and $P(B) = \frac{1}{5}$. Find:
   a. $P(A \cap B)$
   b. $P(A \cap B')$
   c. $P(A' \cap B')$

4. An urn contains 6 white and 9 black balls. If 4 balls are to be randomly selected without replacement, what is the probability that the first 2 selected are white and the last 2 black?

5. In the NU Foundation Programme (NUFYP) 36 percent of students like biology, and 22 percent of the students that like biology also like chemistry. In addition, 30 percent of the students like chemistry. What is
   a. the probability that a randomly selected student likes both biology and chemistry;
   b. the probability that a randomly selected student likes biology given that he likes chemistry?

6. Consider 3 urns. Urn A contains 2 white and 4 red balls; urn B contains 8 white and 4 red balls; and urn C contains 1 white and 3 red balls. If 1 ball is selected from each urn, what is the probability that the ball chosen from urn A was white, given that exactly 2 white balls were selected?

7. Ninety-eight percent of all babies survive delivery. However, 15 percent of all births involve Caesarean (C) sections, and when a C section is performed the baby survives 96 percent of the time. If a randomly chosen pregnant woman does not have a C section, what is the probability that her baby survives?

8. When coin A is flipped it comes up heads with probability $\frac{1}{4}$, whereas when coin B is flipped it comes up heads with probability $\frac{3}{4}$. Suppose that one of these coins is randomly chosen and is flipped twice. If both flips land heads, what is the probability that coin B was the one flipped?

9. Events A, B and C are defined in the sample space S such that $P(A) = 0.4$, $P(B) = 0.2$, $P(A \cap C) = 0.04$ and $P(B \cup C) = 0.44$. The events A and B are mutually exclusive and B and C are independent. Draw a Venn diagram to illustrate the relationship between the three events and the sample space. Find:
   a. $P(B|C)$
   b. $P(C)$
   c. $P(B \cap C)$
   d. $P(A' \cap B' \cap C')$
   e. $P(C \cap B')$

10. In a factory, machines A, B, and C produce electronic components. Machine A produces 16% of the components, machine B produces 50% of the components and machine C produces the rest. Some of the components are defective. Machine A produces 4%, machine B 3% and machine C 7% defective components.

Find the probability that a randomly selected component is

- **a.** Produced by machine B and is defective
- **b.** Defective.

Given that a randomly selected component is defective,

- **c.** Find the probability that it was produced by machine B.

11. Jane always goes to work by bus or takes a taxi. If one day she goes to work by bus, the probability she goes to work by taxi the next day is 0.4. If one day she goes to work by taxi, the probability she goes to work by bus the next day is 0.7
Given that Jane takes the bus to work on Monday, find the probability that she takes a taxi to work on Wednesday.

12. Consider an urn containing 12 balls of which 8 are white. A sample of size 4 is to be drawn with replacement. What is the conditional probability that the first and third balls drawn will be white, given that the sample drawn contains exactly 3 white balls?

13. The members of a wine tasting club are married couples. For any married couple in the club, the probability that the husband is retired is 0.7 and the probability that the wife is retired is 0.4. Given that the wife is retired, the probability that the husband is retired is 0.8.

For a randomly chosen married couple who are members of the club, find the probability that

- **a.** Both of them are retired,
- **b.** Only one of them is retired,
- **c.** Neither of them is retired.

Two married couples are chosen at random.

- **d.** Find the probability that only one of the two husbands and only one of the two wives is retired.

14. Suppose that we have 3 cards identical in form except that both sides of the first card are coloured red, both sides of the second card are coloured black, and one side of the third card is coloured red and the other side black. The 3 cards are mixed up in a hat, and 1 card is randomly selected and put down on the ground. If the upper side of the chosen card is coloured red, what is the probability that the other side is coloured black?

References:
Some of the questions on this worksheet were reproduced from the following sources;
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